The Elicitation of PLAY in Virtual Reality

Literature Review

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ABSTRACT

This review examines virtual reality (VR) as a tool for emotion elicitation, focusing on its potential to induce the PLAY system from affective neuroscience. While VR reliably enhances emotional intensity, presence, and engagement compared to traditional methods, most studies rely on dimensional or discrete models that fail to capture PLAY as a distinct neurobiological system. Instead, PLAY is typically reduced to general positive affect or excitement. Drawing on Panksepp's framework, PLAY is defined as a subcortical system associated with social joy, interaction, and specific neurochemical mechanisms. A synthesis of VR studies highlights consistent effects of interactivity, multisensory design, and presence, but also reveals methodological limitations, including small sample sizes and reliance on self-report measures. The review identifies a key gap: the absence of frameworks for eliciting and measuring PLAY in VR. It argues for integrating affective neuroscience with VR research through multi-level behavioral, physiological, and subjective approaches.

KEYWORDS

Virtual Reality, Emotion Elicitation, PLAY system, Affect

1 Introduction

Emotions are evolutionarily primary processes of the brain, forming a core aspect of human psychological existence. [19] Affective science is an interdisciplinary field concerned with the study of emotions. Over the past decade, there has been growing interest in the relationship between human cognition and emotion. [27, 42, 61] Affective science has thus become more central to research in cognitive science. To facilitate such research, mood induction procedures (MIPs) have been developed to reliably induce emotions in an experimental setting. [2, 61] Emotion elicitation in research aims to induce temporary and controlled affective states, allowing for precise measurement of their effects. The effectiveness of such elicitation methods is typically evaluated based on success rate (i.e., the proportion of participants who experience the target emotion), the intensity of the induced state, and the avoidance of demand effects (i.e., responses influenced by participants' expectations of the study).[27:670]

Virtual reality (VR) is a combination of hardware and software systems designed to engage users in immersive, computer-generated environments.[29] The interaction loop between the user and the virtual world is VR's fundamental mechanism. This loop involves tracking the user's physical movement, then translating it to a virtual representation of the user (avatar), which then interacts with the environment. The outcome of this interaction is then conveyed back to the user through sensory channels (usually visual and auditory). It is important that this loop happens in real-time to maintain immersion. [29]

Presence occurs when a user becomes so immersed in a virtual environment that they become less aware of the underlying technology. [29] The ability to induce presence—often described as the sense of being there—is a defining characteristic of VR. By enabling users to engage with lifelike environments in a controlled yet realistic manner, VR is particularly well-suited for emotion elicitation. Studies have shown that virtual reality is more effective for emotion induction than traditional MIPs like static image-based methods.[12, 33, 45, 50, 53, 64]

Beyond its use as a mood induction procedure (MIP) in psychological research, the elicitation of PLAY in virtual reality has several practical applications. One important area is in clinical contexts, where it may be used as treatment for certain disorders. A lack of PLAY in childhood has been linked to attention deficit hyperactivity disorders (ADHD) [43, 52]. Rather than having a purely biological dysfunction, Panksepp proposes that many children diagnosed with ADHD may instead have unmet PLAY urges that have been suppressed by rigidly structured activities and early academic pressures.[40]

Evidence suggests that PLAY may alleviate ADHD symptoms, as it promotes neuroplastic changes in the frontal cortex and amygdala—supporting "pro-social" circuit development—unlike psychostimulants, which actually suppress PLAY. [40] It may also have therapeutic potential for depression by fostering social bonding, laughter, and positive affect, with supporting evidence from animal models [52]. More broadly, VR environments can enhance mood and reduce psychological stress. PLAY elicitation in VR can also enhance entertainment experiences, with research showing that VR gaming produces more emotional responses than desktop displays [36].

Despite its potential, there are currently no studies on the elicitation of PLAY in virtual reality. This literature therefore examines emotion elicitation in VR, with a particular focus on emotions that may be mapped to Panksepp's PLAY system.

2 Panksepp's PLAY System

Understanding PLAY is crucial to reviewing the elicitation of PLAY in VR. We first begin with defining the framework that PLAY emerges from.

2.1 Primary Emotional Systems

Jaak Panksepp's affective neuroscience framework offers an evolutionary view of emotions. Rather than being constructed through language, cognition, or culture, Panksepp argues that emotions are objective primary processes that are biologically ingrained in ancient brain regions [38, 41].

Since these processes originate in evolutionarily conserved deep brain regions, the capability to experience feelings is not unique to humans [38, 41]. They are shared across mammals[41, 44], which challenges views that animals are simple stimulus-response machines[39, 44] and supports the use of animal research to better understand human emotions[39, 41].

The framework organizes primary, secondary, and tertiary processes into a hierarchical model of the mind[38, 41, 44]. Tertiary processes involve higher cognition associated with the neocortex, secondary processes involve learning and conditioning, and primary processes are associated with the deepest (subcortical) regions of the brain[38, 41, 44]. Raw emotional experience (or affective consciousness) resides in the primary level of this hierarchy, so it comes before and grounds cognitive experience [38, 41]. So thinking does not create emotions, but helps regulate and interpret them[39, 41].

This implies that consciousness begins with feeling rather than cognition, so the neocortex is not required for consciousness to exist[38, 41]. As mentioned before, the deep brain structures that generate basic feelings and a minimal sense of self[38, 41] are not unique to humans. Simple organisms without complex reasoning or language may therefore have a basic form of consciousness based on affective experience[38, 39].

In Panksepp's framework, emotions are primary processes that both guide behaviour and generate subjective feelings[39, 44]. They give experiences their value, or valence: positive feelings encourage approach behaviour, while negative feelings promote avoidance[38, 41]. Without these systems, learning would lack direction, and behaviour would not be meaningfully organised[38, 41]. They therefore play a crucial role in survival by preparing organisms to respond to their environment[39, 41]. As a result, emotional systems form the foundation of motivation, personality, and mental life[39, 41].

Panksepp identifies at least seven core affective systems: SEEKING, FEAR, RAGE, LUST, CARE, PANIC/GRIEF, and PLAY. These are distinct but interacting subcortical networks[38, 39, 41, 44]. The use of capital letters highlights that these are scientific terms referring to brain systems, rather than everyday emotional labels[41]. The activation of each system produces coordinated behavioural, bodily, and feeling responses[41, 44]. Although these systems are biologically hardwired through evolution, they can be shaped by experience and form the basis for more complex human emotions[39, 41]. Panksepp critiques several influential but limiting perspectives. William James's peripheral theory, which proposes that emotions are the perception of bodily changes, ignores deep brain processes that generate emotional experiences[41]. Panksepp argues that scientific blind spots persist today due to Skinnerian or Tinbergen/LeDoux behaviorism views on emotions, which prevent investigation of primary-process affects or animal feelings in neuroscience studies [41] [38] Views that assume that primary-process emotions are unconscious and only higher cognitive emotions become conscious fail to explain how subcortical activities become translated into feelings and ignore evidence that decorticated animals retain rich affective lives. [41]

Evidence from pharmacology and cross-species brain imaging consistently points to subcortical circuits as generators of affective experience[41]. There is evidence that electric brain stimulation at specific subscortical sites produce emotional behaviors and affective experiences, and that predictions in animal research are validated when translated to human research[41].

This framework has important implications for understanding mental health. Disorders such as depression and anxiety can be seen as disruptions in core emotional systems, rather than purely cognitive problems[38, 41]. This means that effective treatment should consider these underlying affective systems, not just thoughts and beliefs[39]. More broadly, affective neuroscience helps reconnect emotion with biology and evolution, addressing earlier approaches that overlooked subjective experience[38, 39].

2.2 What is PLAY?

PLAY is one of the primary-process emotional systems identified by Panksepp. It facilitates social joy—a positive feeling that drives "rough-and-tumble" social behavior. Panksepp emphasizes that this is not just a behavior learnt through imitation or experience, but an innate urge that is genetically determined[42]. For example, rats raised in total isolation from birth still play normally within seconds of encountering another rat for the first time[42]. Play emerges in mid-adolescence, peaks during youth, and diminishes as animals go through puberty[42]. Large frontal lesions in adults increase playfulness[42], suggesting higher brain regions suppress PLAY as an ancient urge.

Touch is identified as the main sensory system for PLAY[42]. Rats, for example, have skin around the nape and shoulders that arouses PLAY when stimulated, similar to ticklish areas at the back of the neck and ribcage in humans[42]. The affective quality of touch is processed by the nonspecific thalamic nuclei (especially the perifascicular area) [42, 43, 52], which supports play motivation. Importantly, this happens outside of neocortical cognitive pathways, as the neocortex is not necessary for PLAY. Animals with their neocortex removed still engage in abundant play, indicating that PLAY is subcortically organized and constitutes a primary process emotion. Other brain structures key to PLAY include the ventral striatum/nucleus accumbens[38, 52] (which mediates the rewarding aspects) and the brainstem dopamine systems[38, 52] (SEEKING-related circuits that provide the euphoric arousal accompanying play).

Play does not require the neocortex to occur, but it does powerfully shape neocortical development. It activates neural growth factors in the frontal cortex[42] (an area involved in decision making, self-control, and social behavior) and the

amygdala[42] (which helps process emotions). Play also encourages the formation of new neural connections[42], especially in the hippocampus[42] (a region important for memory and learning). Through epigenetic changes (long-term modifications of gene expression patterns), play strengthens prosocial neural pathways, refines executive functions, and builds social intelligence[42]. Panksepp and Biven suggest that play may work with REM sleep/dreaming to help organize emotional and cognitive information during waking life[42].

Play is regulated by several neurochemicals that shape how rewarding and flexible it feels. Endogenous opioids support positive social emotions, and their levels can even predict play outcomes[42, 43, 52]. Cannabinoids also promote play, contributing to its "giggly" quality[42, 52]. Dopamine drives the motivation and reward of play, but too much can suppress it by making behavior more rigid[42, 52]. Other chemicals, such as acetylcholine, norepinephrine, and serotonin, have broader effects, with the latter two generally inhibiting play[42, 43, 52]. Play also increases IGF-1, a growth factor that supports brain development and has anti-depressant effects[42].

PLAY can be confused with exploratory curiosity driven by the SEEKING system. While SEEKING (the dopamine-fueled urge to explore) participates in PLAY (especially anticipatory excitement), the two are distinct. For example, dopamine-activating chemicals like amphetamines increase SEEKING but reduce PLAY[42].

PLAY is also not RAGE. PLAY involves rapid role reversals and joyous vocalizations (50-kHz calls in rats), while aggression features defensive postures and complaint vocalizations (22-kHz calls in rats) [42]. Testosterone promotes adult fighting but reduces PLAY in juveniles as PLAY devolves into real aggression[42].

2.2 Behavioral Indicators of PLAY

The rough-and-tumble interactions that characterize play (e.g. chasing, sparring, play-fighting, and wrestling) are accompanied by rapid role reversals between dominant and submissive positions[43]. Play operates by a reciprocity rule. If a stronger animal dominates all interactions, the weaker partner stops responding to play engagements. "Winners" must handicap themselves to maintain play[42]. Nape contacts[52], pinning behavior (in which one animal lies on its back while the other assumes a seemingly dominant posture), poking and nipping, pouncing, tumbling, and circular chasing patterns where participants continuously compete for advantageous positions while maintaining engagement further illustrate the dynamic and reciprocal structure of play[43].

Young animals will show a strong drive to engage in play as they experience it as inherently rewarding. Experimental evidence shows that rats will actively seek play opportunities by traversing mazes or showing preferences for environments associated with previous play experiences[52].

Laughter is an indicator of PLAY. Parallels have been drawn between rat 50-kHz ultrasonic vocalizations (which occur both during play and anticipation of it) and human laughter[52]. Humans may also exhibit "dark laughter"–laughing at other's misfortunes, though this reflects higher cognitive functions and is not intrinsic to PLAY [42].

While "you know it when you see it," Panksepp and Biven adopt Gordon Burghardt's five criteria to recognizing play[42]. All five of these must be satisfied simultaneously. The first is behavior whose adaptive functions are not immediately evident[3]. PLAY is not functionless (there are numerous long-term-evolutionary and short-term benefits of PLAY) but instead appears "not fully functional" while it is happening. The second is that play is spontaneous, done for its own sake, pleasurable, voluntary, or rewarding[3]. The third is that play imitates real adult behaviors or activities, but does so in a way that is exaggerated, awkward, incomplete, or modified. The fourth is that play involves repeatedly doing similar things but with constant variation, which distinguishes play from exploration (as, for example, play persists even after a new object becomes familiar) [3]. The fifth is that animals must be well-fed, safe, comfortable, and healthy for play to occur, as stressors reduce play[3].

These behavioral indicators assist in detecting the arousal of the PLAY system; therefore, they are important to consider when determining whether VR can elicit PLAY.

3 Mapping PLAY onto Common Emotion Frameworks

Studies on the elicitation of PLAY in VR are lacking, with most research instead relying on dimensional models of emotion. Panksepp does not treat valence, arousal, and dominance as abstract descriptive dimensions[44]. He views arousal as the intensity of activation of the core emotional systems[44]. Panksepp and Watt acknowledge that primary-process emotions like PLAY underpin the "varieties of emotional valence, arousal, and power" observed at higher cognitive levels[44]. Accordingly, a loose mapping between Panksepp's framework and dimensional models can be constructed, allowing for the identification and analysis of studies that elicit emotions related to PLAY.

The PANAS scales measures Positive Affect (PA) and Negative Affect (NA) as two largely independent dimensions[60]. PLAY clearly aligns with high Positive Affect due to the high energy and high engagement[60]. Panksepp also states that the PLAY system generates positively valanced states[44]. Russell's circumplex organizes affect along two bipolar dimensions valence (pleasure-displeasure) and arousal[51]. Here, PLAY maps to high pleasure (positive valence) and moderate-to-high arousal, where states like excitement enthusiasm sit [51].

Framework	PLAY's Location/Equivalent	Characteristics
Panksepp (basic Emotions)	Primary-process PLAY system	Subcortical circuit; dopamine/opioid mediated; social joy; laughter-like vocalizations
Russell (Circumplex)	Northeast quadrant — high pleasure,	Aligns with excitement,

	moderate-high arousal	enthusiasm; distinct from pure pleasure (contentment) or pure arousal
Watson & Clark (PANAS)	High Positive Affect (PA)	"Excited," "enthusiastic," "active"; not just absence of negative, but positive engagement
IAPS/SA-APS (image-based)	Likely high-arousal positive images	Cross-cultural studies show valence correlates highly, though arousal ratings vary [32]

While dimensional models capture the subjective experiential qualities of the PLAY system (such as valence and arousal), Panksepp's framework operates at a primary level of analysis that embodies the PLAY system with neurobiological and behavioral qualities that cannot be reduced to generalized affective dimensions[44]. Dimensional models typically examine emotions at a tertiary (cognitive) level[44]. As mentioned earlier, Panksepp is critical of such approaches as they ignore neuro-evolutionary evidence [37]. Clinical insights that link PLAY and ADHD emerge from understanding PLAY as a specific neurobiological system, rather than as generic positive affect. Although cognition and emotion are closely integrated at tertiary levels, primary-process emotional systems operate independently of higher-order cognition.

Panksepp argues that valence-arousal circumplex models are insufficient because they reduce emotions to broad descriptive characteristics without addressing their underlying neural organization[44]. For example, primary emotional systems exhibit distinct neurodynamic and neurochemical specificities, including the roles of opioids, cannabinoids, and specific dopamine dynamics that cannot be captured by a generic arousal axis with no chemical specificity[44]. Rather than falling on this axis, each emotion also has distinct characteristic arousal qualities[44]. These systems are further distinguished by specific behavioral signatures (e.g., rough-and-tumble play, rhythmic movement patterns) and social communicative signals (e.g, 50kHz ultrasonic vocalisations in rats) [38].

Furthermore, the developmental specificity of PLAY Is lost when reduced to valence and arousal, as PLAY is a primarily juvenile phenomenon tied to critical periods of social development and neocortical maturation[44]. Russell's two-dimensional model (pleasure-arousal) cannot distinguish between the non-social excitement of discovery/anticipation (SEEKING) and the social joy of rough-and-tumble play (PLAY). Both would map to high pleasure and high arousal, but they are neurobiologically distinct[44]: SEEKING is associated with foraging, exploration, and appetitive motivation, whereas PLAY is linked to social bonding and joy[52].

Russell's circumplex model explicitly positions itself as a framework for describing the cognitive structure laypeople use to conceptualize affect, acknowledging that affective terms overlap gradually and do not map onto discrete biological systems[51]. Such models are therefore less effective in representing subcortical emotional specializations[44]. Panksepp's model instead investigates emotions at their evolutionary foundation while still accounting for complexity that emerges when these primal systems interact with human cognitive capacities. [41]. His hierarchical model provides a framework for integrating basic emotion theories with dimensional models [41]. From this perspective, dimensional models such as the circumplex are not rejected outright, but are considered incomplete: they describe how emotions are experienced and conceptualized rather than how they are biologically generated.

Panksepp and Watt argue that differences between basic emotion and dimensional approaches reflect differences in levels of analysis, not fundamental theoretical incompatibility[44]. This suggests the need for integration across primary, secondary, and tertiary processes. To achieve this, Panksepp advocates a "psycho-neuro-ethological triangulation" approach[44]. This approach infers emotional states not solely from subjective report, but through consistent patterns across observable behaviors, neurochemical and neural activity, and functional methods such as place preference and revaluation paradigms. Species-specific signals (e.g. 50kHz vocalizations in rats) are treated as analogues of affective expression, providing objective markers of emotional states. This approach offers a basis for assessing elicited emotions, emphasizing multi-level validation and maintaining scientific rigor through falsifiable predictions.

A table showing VR Elicitations by emotion model can be found in Appendix A. Somarathna et al. (2023) [53] notes that discrete models are preferred when specific emotional states are desired, while dimensional models offer continuous measurement. Meuleman and Rudrauf (2018) [30] argue that emotions like joy involve multi-component processes, including appraisal and action tendencies, which are not captured by simple valence-arousal models.

While no study has explicitly targeted Panksepp's PLAY system, existing research on positive, high-arousal states in VR provides indirect evidence about conditions that may support PLAY elicitation. However, because dimensional models cannot distinguish PLAY from SEEKING, flow, or general excitement, this literature must be interpreted cautiously.

4 Emotion Elicitation in VR

Text, images, and sounds are widely used methods for emotion induction. The international affective picture system, for example, contains thousands of photographs with affective ratings[2]. Such methods are often limited by low ecological validity, reduced immersion, and susceptibility to environmental disturbances[21]. For example, Marín-Morales et al. (2018)[24] argue that these

methods fail to evoke intense emotional responses due to being passive and decontextualized, while Halbig and Latoschik (2021)[14] highlight their inability to simulate real-world interaction dynamics.

In contrast, VR offers a more immersive and interactive experience, enabling users to engage with lifelike environments in a controlled yet realistic manner. Multiple studies demonstrate that immersive VR reliably induces strong positive affect, high arousal, and engagement compared to non-immersive media [7, 14, 19, 20, 30, 33, 46, 53, 54, 56, 58, 65] For example, Niu et al. (2019)[33] conducted a comparative experiment showing that VR elicited higher arousal, more positive valence. Yeo et al. (2020)[65] found that interactive VR significantly improved mood compared to passive VR viewing, with effects mediated by presence. Reviews by Marín-Morales et al. (2018)[24] and Somarathna et al. (2021)[53] conclude that VR consistently produces greater emotional intensity and engagement across studies. Marín-Morales et al. (2018)[24] note that many studies rely heavily on self-report measures, which may not capture the full complexity of emotional states.

Passive 360° videos and static environments can induce positive affect, as demonstrated by Felnhofer et al. (2015) [10]. However, active, interactive tasks produce stronger effects, particularly in terms of arousal and engagement[36, 63]. Such tasks are likely conducive to PLAY elicitation as PLAY is characterized as active and interactive (involved rough-and-tumble play, rather than passive perception of stimuli).

While Marín-Morales et al. (2018)[24] and Tian et al. (2021)[1] show that VR produces stronger physiological signals than traditional media, these measures primarily capture arousal, not discrete emotional systems. Appendix B contains a table of studies comparing VR to non-immersive media. All studies found that VR is better than non-immersive media for intensity and presence. Mostajeran et al., 2021[31] is the exception as it found that photos were better for relaxation. Physiological signals were also stronger in VR. A noted trade-off is simulator sickness.

Appendix C contains a table of studies comparing VR environments to real world environments to test the ecological validity of VR. Both studies listed found that emotional intensity in VR and the real world are similar. Temporal patterns of emotion fading are also similar. VR is thus an ecologically valid.

Prescence

A defining characteristic of virtual reality (VR) is its ability to induce presence. Presence happens when the subject's awareness of the external physical environment and technological interface disappears, and they are fully embodied within a mediated environment[29]. The degree of presence experienced is influenced by several factors, including interactivity and real-time system responsiveness, the quality and integration of multisensory feedback, user engagement, and the ecological validity of the virtual environment [29] Several studies found a bidirectional relationship between presence and emotions: presence is greater in emotionally arousing environments, and emotional arousal is greater when presence is greater[45, 48, 49, 64]. Yeo et al. (2020) [65]found that presence mediated the relationship between interactivity and mood improvement. Bluman et al. (2023)[1] found that interactivity enhances negative mood reduction but requires neutral prior mood to increase joy.

Embodiment (the sense of owning and controlling a virtual body induced by VR avatars that move synchronously with the user's real body[28])has been shown to enhance emotional responses[11, 28]. High embodiment intensifies emotional responses to virtual stimuli (including arousal and positive valence) suggesting a close link between bodily presence in VR and play-like affective states.

However, embodiment is rarely studied in isolation, making it difficult to determine whether its effects are independent or mediated by interactivity and presence.

Social and Collaborative Engagement

VR environments support rich social interaction—both with avatars/agents and real users. Collaborative VR with tasks that require positive interdependence (coordinated action, mutual help) enhance social presence, positive affect, and prosocial motivation. [5, 6, 17, 18, 30, 49, 59].

However, type of social presence matters. A companion who is merely present and potentially evaluative[57] may impair performance and reduce positive affect, while a companion who is a collaborator[62] or a helper [5] enhances positive experience. This suggests that environments that introduce judgment or performance pressure may suppress play, showing that positive social interaction is not just about the presence of others but also the nature of the interaction.

Similarly, the effectiveness of avatars and virtual agents depends on design. Sutskova et al. (2023)[57] found that users are sensitive to the perceived agency of virtual characters, and that they respond differently to human-like minds versus AI.

The literature on virtual characters reveals a tension between realism and engagement. Highly realistic human characters may elicit stronger initial attention[8] but can also create expectations that lead to disappointment [34]. Stylized, non-human character are often rated more positively[8], perhaps because they are clearly artificial and therefore not subject to the "uncanny valley" effect. This suggests that for PLAY elicitation, designers may be better off with clearly non-realistic characters rather than striving for photorealism.

Social and collaborative engagement is relevant to consider as PLAY is characterized by mutual engagement, shared goals, and reciprocal positive feedback. Environments that aim to induce PLAY must be mindful of this

4.1 Measurements

A methodological challenge in PLAY research is how to measure it. Most studies rely on self-report questionnaires, but there is growing interest in physiological measures that can capture the autonomic arousal characteristic of PLAY.

Self-report measures consistently show that VR enhances positive affect and arousal compared to less immersive media. Physiological measures generally support this conclusion, though with more variability. The relationship between self-report and physiological measures is not always straightforward: Pavic et al. (2023)[45] found significant self-report effects but weaker

physiological effects, while Niu et al. (2019)[33] found robust effects in both domains. This may reflect differences in the stimuli (360° videos versus interactive games) or in the timing of measurement. Future research should aim to integrate self-report and physiological measures more systematically, perhaps using continuous physiological recording to capture moment-to-moment changes in arousal during playful interaction.

Meuleman and Rudrauf (2018)[30] took a multi-componential approach, collecting self-report data on appraisal, motivation, physiology, feeling, and regulation after participants played various VR games.

Studies using EEG, skin conductance response (SCR), heart rate variability (HRV), and other physiological measures confirm that immersive VR elicits stronger emotional arousal than non-immersive conditions[19, 24, 25, 58]. Marín-Morales et al. (2018(=)[24], for example, achieved 75% accuracy for arousal classification and 71.21% for valence classification using SVM models with EEG and heart rate dynamics.

Ochs et al. (2022)[35] analyzed multimodal behavioral cues during a medical training scenario where doctors practiced breaking bad news to a virtual patient. Using machine learning on verbal and non-verbal cues, they found they could accurately predict users' sense of presence and copresence, demonstrating that objective behavioral measures can assess subjective virtual experiences.

Appendix E outlines measures used across different studies.

4.2 Design Choices

Design choices made in the construction of a virtual environment has direct effects on the emotion that environment elicits in the user. It is thus important to look at the design elements intentionally considered in studies looking at emotion elicitation in VR and how those elements can potentially elicit PLAY.

Multiple studies converge on four key design categories. These categories provide a generalizable framework for VR scene design, where emotional states are constructed through consistent patterns across dimensions.

Dimension	Characteristics	Affective Impact
Semantic/Intention	Foreground themes, symbolic meanings, background environment, narrative elements	Defines emotional valence (positive/negative)
Sensory (Color, Sound)	Hue, saturation, brightness (identified as the most critical visual emotional element by Luo[21]); rhythm, melody, auditory structure	Affects valence and arousal (musical tempo increases arousal)
Dynamism	Lens conversion, object/camera/user movement and speed, plot arrangement, pacing, scene transitions	movement creates stress
Interaction	Agency, constraints, navigation freedom	Primarily affects dominance; goal obstruction creates frustration/anger

Sources: Luo (2024)[21], Dozio et al. (2022)[9], Marcolin et al. (2021)[23].

Luo (2024)[21] extracted reoccurring emotion features from existing labelled audio, video, and image materials (such as IAPS). From these features, they deduced four categories fundamental to differentiating emotions. Colour was identified as the most critical element, as color is the most important visual element and vision is key to induction in VR. Intention, sound, and dynamic were the other three categories.

Steinhausser et al. (2022) [55] developed and validated design guidelines for emotion-inducing Virtual Environments based on a literature review of video games and real environments. From this, they identified 24 guidelines (12 positive, 12 negative) that were sorted into three central design aspects: themes, colors and textures, and lighting. Similar to Luo (2024)[21], Steinhausser et al. (2022)[55] used algorithmic color analysis to identify emotion-specific color palettes. For example, light blue and yellow were identified as corresponding to joy in both studies. Appendix F provides more examples of studies developing specific systems or design methodologies and elements for emotion elicitation.

Rivu et al. (2022) [50]notes that elicited emotions persist, which can be considered when designing scene transitions in VR, and audio cues can induce emotions without disrupting visual experiences. Studies in Appendix D manipulate design elements more broadly.

Multiple studies that explore elicitation of emotions agree on patterns most consistent in eliciting those emotions. This is useful when considering what design qualities in virtual environments could be useful in eliciting PLAY.

4.3 Individual User Differences

A consistent finding across studies is that not all users respond to VR experiences in the same way. Individual differences in personality, prior experience, and cultural background can significantly affect experience in VR. These findings indicate that emotional responses in VR are not uniform, but shaped by individual differences, which would complicate attempts to design universal PLAY-inducing environments.

Dirin and Laine (2023)[8] found that pet owners were more engaged with the virtual dog in their study than non-pet owners. Jin (2010)[15] found that individuals with higher interdependent self-construal (the tendency to view oneself in relation to others) experienced stronger social presence and interaction with a virtual character, suggesting that users' pre-existing social orientation affects their engagement with virtual social agents. Pavic et al. (2022) discusses ethical considerations, noting that "positive

emotions are not valued and expressed in the same way in different cultures."[46]

5 Limitations and Ethical Considerations

Many studies have small sample sizes. Markowitz and Bailenson (2021)[26], for example, note that studies in their systematic review had a mean of 88.36 participants per study and a positively skewed distribution (median 59). They also note that less than half of the studies measured presence, despite its importance as a mediator of emotional effects. Woodworth and Borst (2024)[63] acknowledge that their sample was limited to young adults (mean age 21.5). This is also a common limitation across the literature: most studies use convenience samples of university students, limiting the generalizability of findings to other populations, including children, older adults, and clinical populations. These methodological limitations point to the need for larger, more rigorous studies in the future.

Woodworth and Borst (2024)[63] also acknowledge a relatively high proportion of males (34 out of 54) in their study, though Panksepp and Biven (2012) highlight that there is no gender differences in the urge to play[42:444].

There have been attempts to create datasets for VR. Li et al. (2017)[19] created public dataset of 73 immersive VR video clips annotated with valence, arousal ratings. Suhaimi et al. (2020)[56] created a 360° video dataset using commercial content (couple relationships, skydive, evil encounters, underwater experiences, puppies) for emotion elicitation.

Dozio et al. (2022)[9], however, address the lack of validated VE sets comparable to IAPS images or IADS sounds. There is a need for more standardized methodologies when it comes to sets of validated virtual environments.

Pavic et al. (2022)[46] raise concerns about the long-term effects of VR use, noting that "there is to our knowledge no literature about addiction to VR experiences, yet it needs to be investigated, especially when setting positive VR interventions." This hints that effective elicitation of PLAY in VR may lead to problematic use.

Cybersickness may directly inhibit PLAY, as physical discomfort conflicts with the felt safety required for play (Burghardt's 5th criterion).

Appendix G summarizes papers that synthesize findings across studies. These identify patterns and gaps and provide recommendations for future research.

6 Conclusion

There is general consensus that VR can elicit diverse emotions, that presence and emotion are interconnected, and that design of virtual environments must match the emotion being elicited. Active and interactive methods have also been shown to produce more ecologically and intense emotions than passive observation. Most studies, however, remain focused on valence and arousal. PLAY can only be implicitly approximated via joy, engagement, or excitement, but it is never explicitly targeted or measured in studies on eliciting emotion in virtual reality. Considering PLAY's status as a distinct primary-process emotional system and thus its potential applications, the absence of its explicit investigation represents a critical gap that future research should address.

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Appendix A

Category	Description	Representative Studies	Notes
Dimensional Models	Emotions represented along continuous dimensions (valence, arousal, dominance)	Riva et al. (2007)[9]; Luo (2024)[21]; Rivu et al. (2021)[50]; Somarathna et al. (2023)[53]; Woodworth & Borst (2024)[63]; Gupta et al. (2025)[13]; Magalhães et al. (2024)[22]; Dozio et al. (2022)4/6/26 4:15:00 AM	Widely used in VR due to compatibility with continuous measurement and physiological signals
Discrete Emotion Models	Emotions treated as distinct categories (joy, fear, sadness, anger, disgust)	Dozio et al. (2022)[53]; Steinhaeusser et al. (2022)[55]; Felnhofer et al. (2015)[10]; Gilpin et al. (2021); Somarathna et al. (2023)4/6/26 4:15:00 AM	Useful for targeted emotion induction but less flexible for nuanced affect
Multi-componential Models	Emotions involve appraisal, physiology, motivation, feeling, regulation	Meuleman & Rudrauf (2021)[53]; Pinilla et al. (2021)[47]; Dozio et al. (2022)[9]; Somarathna et al. (2023)4/6/26 4:15:00 AM	Supports multi-method approaches (triangulation)
Plutchik's Wheel	Emotions structured by intensity gradients	Steinhaeusser et al. (2022)[13]; Gupta et al. (2025)4/6/26 4:15:00 AM	Useful for designing graded VR experiences
Broaden-and-Build Theory	Positive emotions broaden cognition and build long-term resources	Bluman et al. (2023)[1]	Explains functional role of VR-induced positive emotions

Appendix B

Study	Comparison	Sample	Content	Emotions Studied	Measures	Key Finding	Effect Size/Statistics
Niu et al., 2019[33]	VR vs. 2D Display	N=60 (30/condition)	Video clips	Positive/negative emotions, arousal	SAM, PANAS, EDA, ECG, SKT, SSQ	VR > 2D for arousal and positive emotions	PANAS+: F=13.872, p=0.018; EDA peaks: p=0.000; ML accuracy: VR 0.78 vs 2D 0.62
Pallavicini et al., 2019[36]	VR vs. Desktop Gaming	N=24	"Smash Hit" game	Happiness, surprise	VAS, HR, SCR, SUS-II	VR > Desktop for emotion and presence	VAS-HP: η2=0.277; VAS-SP: η2=0.235; SUS-II: VR M=27.1 vs Desktop M=16.42
Pavic et al., 2023[45]	VR HMD vs. 360° on Screen	N=26	360° videos (nature, social)	Positive emotions, arousal	SAM, 8 affects scales, EDA, HR	HMD > screen for valence and arousal	Valence: F(1,25)=13.01, p=0.001, η2p=0.34; Arousal: F(1,25)=5.18, p=0.03, η2p=0.17
Yang and Zhang, 2022[64]	VR vs. 360° vs. 2D Video	N=131 (3 groups)	Journalism stories	Enjoyment, presence, flow, empathy, credibility	Custom scales	VR > 360° > 2D on all measures	Enjoyment: VR M=6.93 vs 360° M=5.09 vs 2D M=3.82; Presence: VR M=6.45 vs 360° M=5.05 vs 2D M=3.65
Gilpin et al., 2021[12]	VR vs. Film vs. Text	N=60 (20/condition)	Pet loss narrative	Sadness (subtypes)	PER, ITC-SOPI, HR, RR, RSA	VR = activating sadness; Film = deactivating	RSA model R2=0.39; Respiration R2=0.29

Mostajeran et al., 2021[31]	360° Video vs. Photo Slideshows	N=34	Forest/urban environments	Mood disturbance, fatigue, stress	POMS, GSR, HR, IPQ	Photos better for arousal reduction	GSR reduction: photos > 360° urban ($p<0.05$)
Riva et al., 2007[49]	VR Anxious vs. Relaxing Environments	—	Anxious and relaxing VEs	Anxiety, relaxation	Presence scales, emotional response	Circular presence-emotion interaction; VR is affective medium	—

Appendix C

Study	Specific Comparison	Sample	Stimuli	Emotions Studied	Measures	Key Finding	Effect Size/Statistics
Rivu et al., 2021[50]	VR vs. Real World	N=39	Audio, video, images, autobiographical recall	Happiness, sadness, anger, excitement	SAM (valence, arousal, dominance)	Comparable results; VR slightly stronger	Excitement (images): $p=0.024$; Happiness valence: median 7 vs 6 (RW)
Chirico and Gaggioli, 2019[4]	VR vs. Natural Environments	—	—	Emotional responses, presence	—	"When Virtual Feels Real"—comparable emotional responses	—

Appendix D

Study	Specific Comparison	Sample	Stimuli	Emotions Studied	Measures	Key Finding	Effect Size/Statistics	Design Elements
Woodworth and Borst, 2024[63]	Active Tasks vs. 360° Video (both in VR)	N=54	4 active tasks vs 7 passive 360° videos	Frustration, boredom, confusion, excitement, pleasure	Custom emotion wheel, SAM	Comparable consistency; tasks wider arousal spread	ICC arousal: 0.963; ICC valence: 0.919	Buzzer Box, Box Sorting, Shape Match, Tai-Chi; "happening to me" feeling
Bluman et al., 2023[1]	Interactive vs. Non-interactive VR	N=124	Terror attack → joy induction park	Joy, negative affect, prior mood effects	PANAS, emotion ratings	Interactivity reduced negative affect; joy increase only in neutral prior mood	—	Interactivity as design principle; prior mood moderates joy induction
Magalhães et al., 2024[22]	Multisensory Combinations in VR	N=80 (35 negative, 45 positive)	Treasure hunt: V+A, V+A+T, V+A+H, V+A+S, V+A+T+H+S	Positive/negative affect (PANAS)	PANAS-PTsv, IPQ, ASQ	V+A+H most effective for positive; V+A+T+H+S for negative	Significant differences across all combinations ($p<0.05$)	Taste, haptics, smell + audiovisual; complexity helps negative emotions
Dirin and Laine,	Virtual Character Design (3 types)	N=50 (12 presenters,	HHVR campus tutorial with 3 VCs	12 "Feelings of Being" (trust,	5-point scales,	Felix the Cat (animal) most liked; realistic	Riitta trust $r=0.46$; Felix trust $r=0.50$	Gamification (hide-and-seek with cat); character

2023[8]		38 spectators)		excitement, frustration, etc.)	interviews	human mixed; fictional human least preferred		familiarity affects trust
Felnhofer et al., 2015[10]	5 Virtual Park Scenarios	N=120 (24/scenario)	Virtual parks (joy, sadness, boredom, anger, anxiety)	Joy, sadness, boredom, anger, anxiety	mDES, IPQ, EDA	4/5 scenarios successfully elicited target emotion	Joy, anger, boredom, anxiety successful; sadness scenario failed (induced boredom)	Lighting + sound: birds→joy, owl→anxiety, construction→anger

Appendix E

Model / Approach	Description	Representative Studies	Measurement Methods	Emotions Studied	Key Insights
Self-Report Only	Subjective emotional experience measured via questionnaires	Yang & Zhang (2022)[1]; Luo (2024)[21]; Kern et al. (2020)[16]; Collange & Guegan (2020)[5]; Bluman et al. (2023)4/6/26 4:15:00 AM	PANAS, SAM, custom Likert scales	Enjoyment, presence, empathy, joy, affect	Larger samples; strong for subjective experience; lacks objective validation
Self-Report + Physiological	Combines subjective reports with bodily signals	Niu et al. (2019)[33]; Pallavicini et al. (2019)[36]; Pavic et al. (2023)[45]; Woodworth & Borst (2024)[63]; Mostajeran et al. (2021)[31]; Gilpin et al. (2021)[12]; Felnhofer et al. (2015)[10]; Magalhães et al. (2024)[22]; Dozio et al. (2022)[9]	Self-report + EDA, HR, ECG, GSR	Arousal, affect, stress, discrete emotions	Most common; enables validation; smaller samples due to equipment
Self-Report + Behavioral / Qualitative	Combines ratings with interviews or observed behavior	Dirin & Laine (2023)[48]; Riches et al. (2019) [48]	Likert scales + interviews/thematic analysis	Social emotions, presence-related feelings	Captures experiential nuance and "felt sense"
Triangulation (multi-method)	Combines self-report, behavioral, and sometimes physiological measures	Rivu et al. (2021)[50]; Steinhaeusser et al. (2022)[55]	SAM + behavioral tasks + exposure methods	Happiness, sadness, fear, joy	Rare but theoretically strongest (aligns with multi-componential models)

Appendix F

Study	Focus	Sample	Stimuli/Method	Emotions Studied	Measures	Key Finding	Effect Size/Statistics	Design Elements

Steinhaeusser et al., 2022[55]	VE Design Guidelines Development	Study I: N=85; Study II: N=41	360° videos and immersive VR (HTC Vive Pro)	Fear, Joy	PANAS-X (Joviality, Fear)	Design guidelines successfully induced target emotions	Significant fear increase in negative VE; joy increase in positive VE	Themes, colors/textures, lighting: dark/red/black for fear; bright/pastel/nature for joy
Dozio et al., 2022[9]	5-Emotion VE Design Methodology	N=48	10 VEs (2 per emotion: happiness, sadness, anger, fear, disgust)	Happiness, sadness, anger, fear, disgust	SAM, BEES, TAS-20	Valence distinguished positive vs. negative; fear most arousing	Fear highest arousal; happiness lowest arousal; anger lower control	Semantic (real-life/symbolic), sensory (visual/auditory/tactile/olfactory), dynamism, interaction
Luo, 2024[21]	VR Emotion Induction System Design	—	8 VR scenes (80 sec each)	Joy, disgust, sadness, fear, calmness	Pleasure/arousal/ dominance model	System designed for 4-quadrant emotion space	Quantified emotion degrees in pleasure-arousal space	Landscape design, plot, audio, lighting; Unreal Engine 4.12; octree algorithm for colors
Collange and Guegan, 2020[5]	VR Gratitude Induction System	Study 1: N=80; Study 2: N=100	Building fire escape with virtual benefactor	Gratitude	Gratitude scales, warmth/competence, social support intentions, IOS	Virtual help induced gratitude and increased social support	Study 1: M=4.13 vs 1.75, $p<0.001$; Study 2: M=4.06 vs 1.84, $p<0.001$; Warmth mediated (B=0.44, $p<0.001$)	Burning building; confederate opened doors; 2-min interaction; no verbal exchange; group belonging manipulation
Kern et al., 2020[16]	Mood Induction by Music/Soundscape	N=51	Virtual Park with treadmill; silence, nature sounds, positive/negative music	Presence, positive/negative affect	Presence ratings, PANAS	Any sound > silence for presence; music tended to induce expected emotions	Significance for sound vs. silence	Audio conditions in VR park; time of day variation (day vs. night)

Appendix G

Study	Focus	Scope	Key Synthesis	Notable Insights
Pavic et al., 2022[46]	Positive Emotions in VR	Various studies (5-1000+ participants)	VR efficient for positive emotions across lifespan; immersion, interactivity, content, sensory modalities are key determinants	Meta-analytic patterns; virtual nature consistently effective; interactive HMD > non-interactive; auditory stimuli enhance

Somarathna et al., 2023[53]	VR for Emotion Elicitation Review	Review of VR environments, videos, games, tasks, avatars, images, 360° panoramas	VR evokes emotions effectively via motivational and empathy mechanisms; ecologically valid paradigm	360° panoramas, games, tasks, avatars evaluated; public datasets; emotion-sensing interfaces reviewed
Markowitz and Bailenson, 2023[26]	VR and Emotion 5-Year Systematic Review (2015-2019)	5-year systematic review	VR used for emotion regulation, as mechanism for social/psychological phenomena, and for understanding emotion theory	Three research types: emotion regulation perspective, emotion as mechanism, emotion theory understanding
Linares-Vargas and Cieza-Mostacero, 2024[20]	Interactive VR Environments and Emotions	16 studies analyzed (Unity: 81.25%, HTC Vive: 37.5%)	Most research uses VR to influence users (56.25%); theories mentioned in only 18.75%	Time range: 1-30 min; lack of theoretical grounding in many studies
Diniz Bernardo et al., 2021[7]	Mood Induction Using VR Review	Recent VR mood induction findings	Synthesis of recent methodologies and effectiveness	Review of procedures and clinical applications
Riches et al., 2019[48]	Factors Affecting Presence in VR Social Environment	N=76, qualitative thematic analysis	8 themes: emotions about self/others, thoughts, physiological reactions, avatar behavior, interactivity, environment	Presence facilitated by genuine responses; decreased by diminished agency, cybersickness